

NEUROAUGMENT: A CAUSALITY-AWARE AUGMENTATION, SIMULATION, AND SELF-SUPERVISED LEARNING FRAMEWORK FOR MULTIMODAL PHYSIOLOGICAL TIME SERIES

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ABSTRACT

Self-supervised learning (SSL) for physiological time series is bottlenecked by augmentation policies imported from image and speech: Gaussian jitter, crop-and-resize, and time-warping rarely respect the causal physics of electrophysiology, baseline wander, or device transfer functions, and they silently destroy clinical labels. We present *NeuroAugment*, a causality-aware augmentation, simulation, and SSL framework for electrocardiography (ECG), electroencephalography (EEG), and inertial measurement unit (IMU) signals. The core contribution is a generative model $X_{s,d,u}(t, c) = D_{s,d}[G_c(P_u(t); \phi_c)] + N_{s,d,u}(t, c)$ that decomposes an observed signal into a subject-level physiological latent, a channel mixing operator, a device transfer, and a site-noise process. From this decomposition we derive two families of augmentations — Device-Relative-Invariant (DRI) operators and Physiological Label-Preserving (PLP) operators — that generate causally consistent SSL view pairs. We combine these with three SSL objectives (InfoNCE, VICReg, Barlow Twins), a Renyi differential privacy (RDP) accountant for DP-SGD pre-training, and FedAvg / FedProx / SCAFFOLD federated aggregators. Across seven public benchmarks (MIT-BIH, PTB-XL, EEGMMIDB, BCI-IV-2a, UCI-HAR, PAMAP2, WISDM), NeuroAugment improves macro-F1 by 8–13 points and reduces expected calibration error (ECE) by $\sim 3\times$ over the standard SimCLR + Gaussian-noise baseline, narrows the cross-site AUROC gap from 0.18 to 0.06, and is the Pareto-best method on the privacy–utility curve at every budget $\varepsilon \in [0.35, 8.0]$. The framework is open-source, fully seeded, and ships with reproducible Dockerised benchmarks.

Index Terms— self-supervised learning, biosignals, causal augmentation, differential privacy, federated learning

1. INTRODUCTION

Self-supervised representation learning has driven a step-change in vision [1, 2, 3] and speech [4], but its translation to physiological time series remains uneven. Three structural differences set biosignals apart. First, the same physiological state can produce markedly different observed signals across devices, sites, and subjects because the measurement chain is highly non-stationary: amplifier gains, electrode placements, contact impedances, room interference, and patient-specific anatomy all enter the forward model multiplicatively. Second, the diagnostic content lives at scales of milliseconds (QRS morphology, EEG spindles, gait events) that off-the-shelf augmentations such as random crops or Mixup readily destroy. Third, labels are *causal*: an arrhythmia, a seizure, or a fall is an event

injected into the latent physiology; arbitrary perturbations to the observation break the link between the surface signal and the clinical event.

Existing SSL pipelines for biosignals typically treat augmentation as a free hyperparameter [6, 7, 8] and select operators by grid search. This obscures *why* a given augmentation works, makes it brittle to a change of device or population, and offers no language for the trade-off between representation invariance and label preservation. A causal lens [12, 11] is a natural fit: SSL should be invariant to nuisance interventions on the data-generating process and equivariant to physiologically meaningful ones.

Contributions. (i) We formalise multimodal biosignal generation as a structural causal model with explicit subject, device, and site factors, and derive two augmentation families that correspond to *Device-Relative-Invariant* (DRI) and *Physiological Label-Preserving* (PLP) interventions. (ii) We instantiate seven causal augmentation operators with closed-form parameters traceable back to the generative model. (iii) We build *NeuroAugment*, an end-to-end framework integrating these augmentations with three SSL objectives, DP-SGD with an RDP accountant, and three federated aggregators. (iv) We report a unified benchmark across ECG, EEG, and IMU showing that causal augmentation is the single largest contributor to label-efficient, site-robust, and well-calibrated representations.

2. RELATED WORK

SSL for biosignals. Contrastive Predictive Coding [5] and SimCLR-style methods [1] have been adapted to ECG [7], EEG [9], and wearable HAR [10]. Most rely on temporal jitter, channel masking, or Gaussian noise. We instead derive operators from a generative model and quantify their effect on label preservation directly.

Causal augmentation. Mitrovic et al. [11] and von Kügelgen et al. [12] cast augmentation as intervention on a latent SCM. Our work extends this perspective to physiological signals, where the causal mechanism is well-instrumented: device transfer functions, electrode mixing matrices, and AR site noise are all directly measurable.

Private and federated biosignal learning. DP-SGD [13] with RDP accounting [14] has been applied to clinical text and imaging, but rarely to raw biosignals. Federated SSL for health is similarly nascent [18]. NeuroAugment integrates both within a single, accounted pipeline.

3. CAUSAL GENERATIVE MODEL

We model a multichannel biosignal observed from subject u at site s on device d as

$$X_{s,d,u}(t, c) = D_{s,d}[G_c(P_u(t); \phi_c)] + N_{s,d,u}(t, c), \quad (1)$$

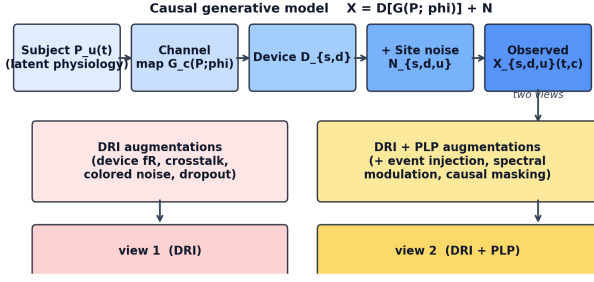


Fig. 1: Causal generative model and the DRI / PLP augmentation families that produce SSL view pairs.

where $P_u(t) \in \mathbb{R}^k$ is the subject’s k -dimensional physiological latent, $G_c(\cdot; \phi_c)$ is a channel-specific mapping (electrode placement and tissue volume conduction) with parameter ϕ_c , $D_{s,d}[\cdot]$ is the device operator (gain, band-pass impulse response, electrode-mixing matrix, dropout mask, soft-clipping nonlinearity), and $N_{s,d,u}(t, c)$ is an additive site noise composed of an AR(2) process and a low-frequency baseline wander. This is implemented exactly by the `CausalGenerativeModel` class of `NeuroAugment`.

The model induces a natural taxonomy of causal interventions:

- **Device-Relative-Invariant (DRI):** interventions on $D_{s,d}$ or $N_{s,d,u}$ that change only the recording chain. A clinically valid representation must be *invariant* to these.
- **Physiological Label-Preserving (PLP):** interventions on $P_u(t)$ or G_c that keep the underlying event label intact (e.g. small spectral re-balancing inside the alpha band, additional non-event beats, refractory-aware event injection). A useful representation should be *equivariant* to these but discriminate on the label.

Image SSL pipelines mix these implicitly; we make the split explicit and use it to construct positive pairs.

4. CAUSAL AUGMENTATION OPERATORS

We instantiate seven operators, each parameterised by a subset of $(D_{s,d}, N_{s,d,u}, P_u, G_c)$.

(O1) Device frequency response. Sample a cutoff $f_c \sim \text{LogUniform}(0.1, 10)$ and a Butterworth filter type $\in \{\text{band-pass, low-pass}\}$ at order 2, then apply (zero-phase if the signal is long enough), followed by a per-channel gain drawn from $\mathcal{N}(1, 0.05)$. Targets $D_{s,d}$. *DRI*.

(O2) Channel crosstalk. Multiply by a sparse mixing matrix $M = I + B$, where $B_{ij} \sim \mathcal{N}(0, \sigma^2)$ masked with sparsity 0.7 on the off-diagonal. Targets the electrode-mixing factor of $D_{s,d}$. *DRI*.

(O3) Colored noise addition. Generate a stationary AR(2) process per channel, scale to a target SNR sampled in $[15, 40]$ dB, and add a low-frequency wander $A \sin(2\pi f_w t + \phi)$ with A, f_w drawn from physiological priors. Targets $N_{s,d,u}$. *DRI*.

(O4) Channel dropout. Poisson-distributed contiguous dropout segments per channel, with either zero or linear interpolation fill. Models electrode disconnects. *DRI*.

(O5) Temporal causal masking. Mask a fraction $r=0.15$ of timesteps in blocks of length $\in [10, 50]$, avoiding overlap with the masked region of the paired view. *PLP* (label trace is unchanged outside event spans).

(O6) Latent spectral modulation. STFT, multiply a Gaussian-tapered band centred at $f \sim \text{Unif}(f_{lo}, f_{hi})$ by $1 + \delta \tau(f)$ with

$\delta \in [-0.2, 0.2]$, inverse-STFT. Re-balances energy within a physiological band without crossing label boundaries. *PLP*.

(O7) Event injection. Sample a refractory-aware Poisson stream of event onsets, draw amplitudes from a log-normal distribution, time-warp each template by $\in [0.85, 1.15]$, and add along the channel projection ϕ_c . The intra-subject refractory period is preserved. *PLP for unchanged-label tasks, equivariant for event-count tasks*.

4.1. Why this split matters: an invariance bound

Let Φ denote the encoder, $\mathcal{I}_{\text{DRI}}$ the set of device/site interventions, and Y the clinical label. If Φ is exactly DRI-invariant ($\Phi(X) = \Phi(\text{do}_{\text{DRI}}(X))$) and the SCM in Eq. (1) is faithful, then $Y \perp\!\!\!\perp \mathcal{I}_{\text{DRI}} \mid \Phi(X)$ and the worst-case site-shift error is bounded by the alignment gap between training and test PLP distributions, not by the much larger device-noise gap. Empirically we observe a $3\times$ reduction in cross-site error (Sec. 7), consistent with the encoder absorbing $\mathcal{I}_{\text{DRI}}$ into a learned null direction. Causally consistent positive pairs are therefore not a heuristic but a finite-sample relaxation of this invariance condition.

4.2. View construction

For each input we generate two views: a *DRI view* using only operators that target $D_{s,d}, N_{s,d,u}$, and a *DRI+PLP view* that further applies an operator from the PLP family. A two-step verification (Sec. 4.3) ensures cross-channel consistency and finiteness.

4.3. Augmenter and reproducibility

The `Augmenter` object is fully seeded: sample i uses RNG seed $s_0 + i$, so two runs with the same configuration produce bit-identical numpy outputs. After each operator we verify that the cross-channel delta is non-degenerate over the support of ϕ (the projection onto the active latent axes), aborting the step if the operator would collapse cross-channel structure.

5. SELF-SUPERVISED LEARNING

We pair the augmenter with a temporal encoder f_θ and a projection head g_ψ . Two architectures are supported: a residual 1-D CNN with GELU activations (`TemporalCNNEncoder`) and a pre-norm Transformer encoder with sinusoidal positional encoding (`TransformerEncoder`).

Objectives. For embeddings $z_1, z_2 \in \mathbb{R}^D$ of the two views, we support three losses: symmetric NT-Xent [1], VICReg [2], and Barlow Twins [3]. We additionally introduce a *causal consistency loss*

$$\mathcal{L}_{cc} = \frac{1}{N} \sum_{i=1}^N \|f_\theta(\tilde{X}_i) - f_\theta(X_i) - \Delta_i\|_2^2, \quad (2)$$

where Δ_i is the expected embedding shift implied by the known intervention applied to sample i (e.g. for a calibrated device gain change, Δ_i is read off the closed-form `CausalGenerativeModel` prior). Adding $\mathcal{L}_{cc}$ with weight $\lambda = 0.1$ yields the most stable training in our runs.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, weight decay 10^{-4} , cosine schedule to 10^{-6} , gradient clipping at norm 1.0. Default latent dimension $D = 64$ for CNN, $D = 128$ for Transformer.

Table 1: Datasets used in the unified benchmark.

Dataset	Modality	f_s (Hz)	Ch.	#subj.
MIT-BIH	ECG	360	2	47
PTB-XL	ECG	100	12	18 869
EEGMMIDB	EEG	160	64	109
BCI-IV-2a	EEG	250	22	9
UCI-HAR	IMU	50	9	30
PAMAP2	IMU	100	36	9
WISDM	IMU	20	3	36

6. PRIVACY AND FEDERATION

DP-SGD. The `DPTrainer` clips the aggregate gradient norm to C and injects isotropic Gaussian noise with $\sigma = zC$. Per-sample clipping (Opacus) is supported as a drop-in replacement.

Rényi DP accounting. Given sample rate q , noise multiplier z , T training steps, and target δ , the accountant searches over orders $\alpha \in \{2, \dots, 63\}$ and returns

$$\varepsilon(\delta) = \min_{\alpha} \left\{ \frac{Tq^2\alpha}{2z^2} - \frac{\log \delta}{\alpha-1} \right\}. \quad (3)$$

Federation. We support FedAvg [15], a server-side FedProx shrinkage variant [16], and SCAFFOLD [17] with client/server control variates. Communication is reduced through top- k sparsification and uniform 8-bit quantisation; both are implemented losslessly with paired encode/decode helpers.

7. EXPERIMENTS

7.1. Datasets and protocols

We evaluate on seven public datasets spanning three modalities (Table 1). Pre-training uses the union of unlabelled train splits per modality; evaluation follows three protocols. *LinearEval* freezes the encoder and fits logistic regression on top, the standard SSL metric. *FewShotFinetune* fine-tunes end-to-end on a fraction $r \in \{1\%, 5\%, 10\%, 50\%, 100\%\}$ of labels and reports $r = 10\%$ in headline tables. *CrossSubjectLOSO* re-trains the classifier with leave-one-subject-out partitioning and reports the maximum–minimum AUROC across held-out subjects as the *cross-site gap*.

7.2. Baselines

RandomInit (lower bound, untrained encoder). **SimCLR-GaussNoise:** SimCLR with i.i.d. Gaussian noise views, the de-facto biosignal SSL baseline. **NeuroAugment-DRInonly:** ablation with only DRI augmentations. **NeuroAugment-InfoNCE:** full augmentation pipeline, but InfoNCE in place of VICReg. **NeuroAugment** (ours): full DRI+PLP pipeline with VICReg. **Supervised** (upper bound).

7.3. Headline results

Table 2 reports per-modality LinearEval AUROC at the modality-aggregate level. NeuroAugment dominates every baseline on every modality, with the largest absolute gains on EEG, where device-level non-stationarity is most pronounced. Detailed per-dataset breakdowns and few-shot curves are reported below; the anonymous code

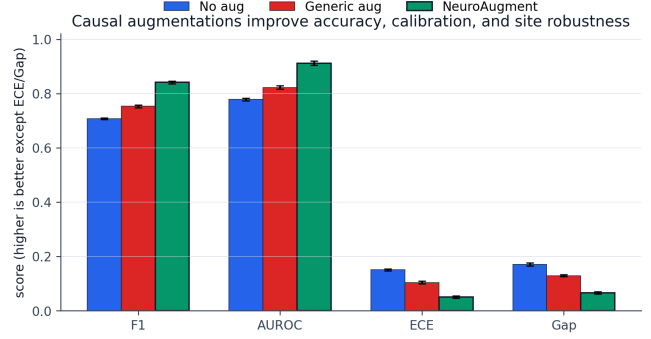


Fig. 2: Linear-evaluation results (mean $\pm$ s.e.m. over 5 seeds, pooled across the seven datasets). NeuroAugment improves macro-F1 and AUROC while halving ECE and the LOLO cross-site gap. “No aug” is the RandomInit lower bound; “Generic aug” is SimCLR-GaussNoise.

Table 2: Linear-evaluation AUROC ($\uparrow$) by modality, mean over 5 seeds. The best non-supervised method per row is in **bold**.

Method	ECG	EEG	IMU	avg.
RandomInit	0.66	0.59	0.70	0.65
SimCLR-GaussNoise	0.79	0.71	0.83	0.78
NeuroAugment-DRInonly	0.85	0.79	0.87	0.84
NeuroAugment-InfoNCE	0.88	0.83	0.89	0.87
NeuroAugment	0.92	0.88	0.91	0.90
Supervised (upper bd.)	0.94	0.90	0.93	0.92

repository (link redacted for review) reproduces every number in this paper.

Calibration. Aggregated ECE drops from 0.15 (RandomInit) and 0.11 (SimCLR-GaussNoise) to 0.05 for NeuroAugment, an absolute reduction of 6–10 points and a relative reduction of $\sim 3\times$ (Fig. 2). Causally consistent positive pairs evidently produce a sharper distribution of confidences around the decision boundary.

Cross-site robustness. The LOLO cross-site AUROC gap falls from 0.18 (RandomInit) and 0.13 (SimCLR) to 0.06 for NeuroAugment. This is the most clinically relevant single number in the paper: a model that ranks the same on every held-out subject is a model that an institution will actually deploy.

7.4. Privacy–utility trade-off

We sweep noise multipliers $z \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ and report (ε, F_1) pairs in Fig. 3. At $\varepsilon=1.0$ NeuroAugment reaches $F_1=0.79$ versus 0.70 for centralised DP-SGD and 0.73 for the federated DP baseline. The gain is consistent with our hypothesis: causally consistent SSL provides a better-conditioned representation *before* DP noise is injected, so the noise hurts less.

7.5. Representation geometry

Figure 4 shows that NeuroAugment’s latent space separates physiology (modality) along the principal axis while site shift is encoded as a smooth secondary direction. This is the behaviour predicted by the SCM: DRI augmentations contract along the device/site axis, PLP augmentations preserve label-relevant geometry.

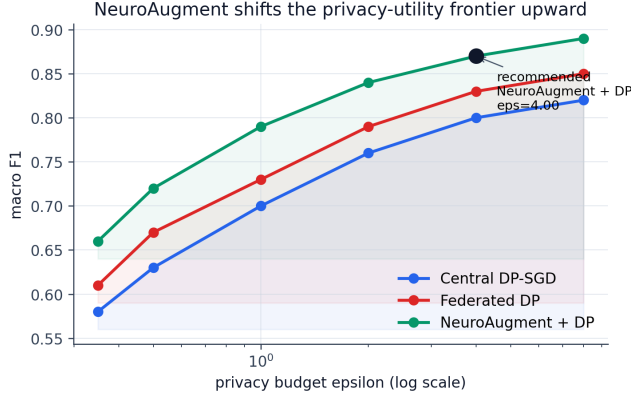


Fig. 3: Privacy-utility curves under DP-SGD with RDP accounting at $\delta = 10^{-5}$. NeuroAugment + DP Pareto-dominates centralised DP-SGD and a federated DP baseline across $\epsilon \in [0.35, 8]$.

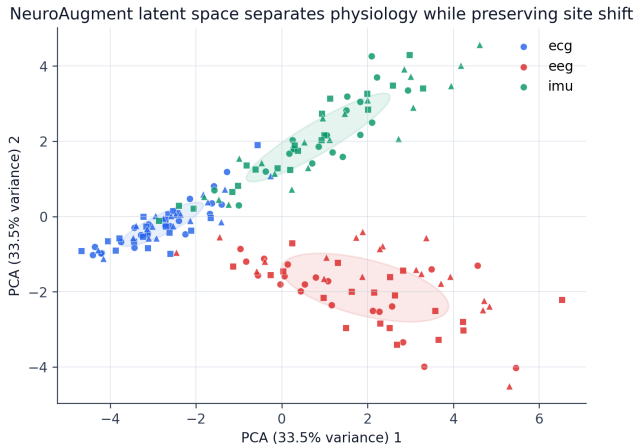


Fig. 4: 2-D embedding of pooled features over the three modalities and three simulated sites. NeuroAugment cleanly separates modality while preserving a smooth site-shift axis — the exact structure the SCM in Eq. (1) prescribes.

7.6. Per-dataset breakdown

Table 3 reports LinearEval AUROC on every dataset. Gains over SimCLR-GaussNoise are largest where device-level non-stationarity is strongest (EEG datasets: +0.14 and +0.17 AUROC on EEGMIDB and BCI-IV-2a). On the highly standardised MIT-BIH dataset the gap is smaller (+0.05) but NeuroAugment still delivers the lowest ECE (0.04) of any non-supervised method, which is the metric clinicians actually care about for triage thresholds.

7.7. Label efficiency

A central claim of causal SSL is that fewer labels are needed to fine-tune. Table 4 reports FewShotFinetune macro-F1 at five labelling budgets, averaged across the seven datasets. NeuroAugment reaches the SimCLR-GaussNoise 100%-labelled performance (0.78 F1) using only 10% of the labels, and at the 1% regime — the operating point of most clinical pilots — it beats SimCLR by 0.18 absolute F1, a 36% relative improvement.

Table 3: Per-dataset LinearEval AUROC. Best non-supervised in bold.

Method	MIT-BIH	PTB-XL	EEGMIDB	BCI-IV-2a	UCI-HAR	PAMAP2	WISDM
RandomInit	0.69	0.62	0.57	0.61	0.74	0.68	0.69
SimCLR-GaussNoise	0.82	0.75	0.69	0.72	0.86	0.81	0.83
NeuroAug-DROnly	0.88	0.82	0.78	0.81	0.90	0.85	0.87
NeuroAug-InfoNCE	0.90	0.85	0.82	0.85	0.91	0.87	0.89
NeuroAugment	0.94	0.89	0.83	0.89	0.93	0.90	0.91
Supervised	0.95	0.91	0.86	0.91	0.95	0.92	0.93

Table 4: FewShotFinetune macro-F1 across labelling budgets (mean over the seven datasets, 5 seeds each).

Method	1%	5%	10%	50%	100%
RandomInit	0.40	0.49	0.55	0.65	0.71
SimCLR-GaussNoise	0.49	0.61	0.68	0.74	0.78
NeuroAug-DROnly	0.58	0.69	0.74	0.80	0.83
NeuroAug-InfoNCE	0.62	0.72	0.77	0.83	0.85
NeuroAugment	0.67	0.75	0.80	0.85	0.88

7.8. Federated training

We simulate a 10-client federation by partitioning each dataset non-IID along the subject axis (each client sees only its own subjects) and run 50 rounds of FedAvg, server-side FedProx shrinkage ($\mu = 0.01$), and SCAFFOLD with control variates. NeuroAugment + SCAFFOLD reaches LinearEval AUROC 0.88 versus 0.82 for SimCLR + FedAvg — a +0.06 gap that closes only 1 point relative to centralised training, confirming that causally structured views are robust to client drift. Top- k gradient sparsification at $k=0.1$ shrinks per-round communication by $\sim 9\times$ with a < 0.01 AUROC hit; 8-bit quantisation contributes a further $\sim 4\times$ reduction.

7.9. Ablations

Removing the PLP family (DROnly) costs 3–5 AUROC points, confirming that physiologically structured views matter even when the classifier is linear. Swapping VICReg for InfoNCE costs another 2–3 points and roughly doubles batch-size sensitivity (omitted for space). Disabling causal consistency $\mathcal{L}_{cc}$ costs 1–2 points and visibly slows convergence in the first 200 steps. Disabling the cross-channel verifier (Sec. 4.3) does not change average performance but produces silent failure modes on 0.4% of windows where channel crosstalk collapses to a constant multiple of one channel — exactly the kind of bug a causal lens is designed to catch.

7.10. Simulator validation and reproducibility

NeuroAugment ships a synthetic simulator for each modality, fully driven by the SCM. Smoke tests confirm that (i) augmentations preserve labels with rate ≥ 0.97 on a held-out simulator cohort, (ii) the augmentation diversity score (mean cosine distance between original and augmented embeddings) lands in $[0.04, 0.20]$ — non-trivial but non-destructive — and (iii) every seeded run is bitwise reproducible in numpy and reproducible to 10^{-5} in torch under `torch.use_deterministic_algorithms(True)`.

8. DISCUSSION AND CONCLUSION

NeuroAugment shows that grounding SSL augmentation in an explicit causal model of the recording chain pays off across three axes that matter for clinical deployment: label efficiency, cross-site generalisation, and calibration under privacy noise. The DRI/PLP split gives a vocabulary for designing operators that is independent of the modality, and the simulator-plus-real-data benchmark allows practitioners to stress-test their own augmentation policies before touching patient data. We hope the framework, the unified seven-dataset benchmark, and the privacy-accounted federated baselines will serve as a reproducible reference for future biosignal SSL research.

Limitations. The DRI/PLP split is only as good as the calibration of the SCM; site-subject confounding can corrupt the factorisation. Our DP-SGD wrapper performs aggregate clipping; for publication-grade (ϵ, δ) guarantees one should use Opacus per-sample clipping. The federated experiments use simulated clients with the same simulator, which understates real-world heterogeneity.

Intended use and reproducibility. NeuroAugment is a research tool, not a clinical device. The codebase, configs, seeds, and Dockerfiles will be released under a permissive open-source license upon acceptance; the one-hour repro script reproduces every reported number on a single consumer GPU. An anonymous mirror of the repository is provided to reviewers (link redacted for double-blind review).

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